

PID Control

Root Locus

Temperature Contol

Jhoberg Quevedo

Engineer

Overshot=0.001

Tr=0.001

erss=0.01

Source Code Octave

PID control

```
%# Control PID rootlocus
%# by Jhoberg Quevedo Ruiz
%# Engineer
```

```
%      u  +      +---+      +-----+y
%      ----->(+)----->| Gs |----->| Gp |-----+----->
%              ^ -      +---+      +-----+      |
%              |              |
%              +-----+-----+-----+-----+
```

```
pkg load symbolic;
pkg load control;
pkg load signal;
```

```
syms S
syms Kp
syms Kd
syms Ki
syms E
syms wn
syms t
```

```
Kpow=49.5
s=tf('s')
```

```
Gpt=Kpow*exp(-t/14.940)+19
Gps=Kpow*exp(-t/14.940)+19
Gp=laplace(Gpt)
Gp=simplify(Gp)
GP=Kpow*-((S-(2/0.3))/(S+(2/0.3)))/(0.53*S+1)
Gptf=(102339*s + 1900)/(2*s*(747*s + 50))
```

$$GPTF = (102339 \cdot S + 1900) / (2 \cdot S \cdot (747 \cdot S + 50))$$

```

figure(1)
hold off
t1 = linspace(0, 100, 1000); % vector time
y1=Kpow*exp(-t1/14.940)+19
plot(t1,y1)
title ("Step Temperature");
xlabel ("Time");
ylabel ("Temperature");
%text (pi, 0.7, "arbitrary text");
legend ("Signal temperature To");

%timeTs=0.1
%E=5%
%Td=0%

[numGp, denGp] = numden(sym(Gp))
numGp=expand(numGp)
denGp=expand(denGp)
den=double(coeffs(denGp))
num=double(coeffs(numGp))

Gp1=tf(Gptf)
[z,p,k] = tf2zp(num,den)
[p,z]=pzmap(Gp1)

%set figure
h = findobj(gca, 'type', 'line');
set(h, 'markersize', 9)
text(real(roots(num)) - 0.1, imag(roots(num)) + 0.1, 'Zero')
text(real(roots(den)) - 0.1, imag(roots(den)) + 0.1, 'Pole')
axis equal

figure(2)

step(Gp1);
title ("Step Temperature");
xlabel ("Time");
ylabel ("Temperature");
%text (pi, 0.7, "arbitrary text");
legend ("Signal temperature To");

%step(feedback(Gp1))
%title ("Step Temperature");
%xlabel ("Time");
%ylabel ("Temperature");

```

```
%text (pi, 0.n7, "arbitrary text");
legend ("Signal Temperature To");
hold off
```

```
%%%%%%%%%%Control PID
```

```
%error state stable rampa unitaria <=0,001
%sobrepaso max<=0.01%
%tiempo levantamiento tr<=0,001 seg
%tiempo asentamiento ts <=0,007 seg
```

```
%calculo PID controller
%1) error estado estable ess
%Ke=limS->0 Gol(S), entrada escalon unitario
limit(GP, S, 0)
Ke=49.5
```

```
%ess=1/(1+Ke)
solve(0.001==1/(1+Ki),Ki)
%Kp=100
```

```
%%%%%%%%%
%2)calculo Overshot sistema 2nd order
clear E
syms E
solve(0.001==exp(-pi*E/(1-E*E)^0.5),E)
```

```
E=0.9102;
```

```
%%%%%%%%%
%3) calculo Tiempo retardo tr
% aproxlinea recta , fr=1/(L*C)^0.5=316.22 , tr>>10*1/fr
clear wn
syms wn
tr=(0.8+2.5*E)/(wn)
wn=solve(0.001==(0.8+2.5*E)/wn,wn)
wn=3075.4
```

```
%4) calculo Tiempo asentamiento ts
Ts=4/(wn*E)
Ts=1.4286e-03
```

```
%5)calculo PID , ecuacion caracteristica Close Loop Gcl
%divisor orden 2 , S*S+2*E+wn+wn*wn
```

```
sd1=-E*wn+i*wn*(1-E*E)^0.5
sd2=-E*wn-i*wn*(1-E*E)^0.5
```

```
sd1=-2799.2 + 1273.9i
```

```
%Gc=K*(s+a)*(s+b)/s
```

```
%sum(Z)-sum(P)=-180
% (teta1-teta4)-(teta2-teta3)=-180
teta4=180-atan(1273.9/(2799.2-0))
teta3=180-atan(1273.9/(2799.2-0.01856))
teta3=180-atan(1273.9/(2799.2-0.066934))
%teta1=atan(182.73/(a-134.68))
% solve(atan(42/(a-40))-(179.26+179.18)=-180,a)
%teta1=atan(1766/a-3075.7)
%sum(tetaZ)-sum(tetaP)=-180
a=1273.9/tan(-0.43)+2799.2
a=21.525
```

```
%Koverall=IILP/IILZ
%Kall=L2*L3/L1
% L2=(42*42+38*38)^0.5
% L3=(42*42+39*39)^0.5
% L1=(42*42+(43.66-40)*(43.66-40))^0.5
% Kall=L2*L3/L1
```

```
% |Gc(S)*H(S)|=1
% K*(S+43.66)*1/((S+1)*(S+2))=1
%
%K=((S+1)*(S+2))/(S+43.66)
KHs=(1/GPTF)*(1/(S+21.525))
S=abs(sn1)
K=eval(KHs)
```

```
K = 0.014498
```

```
%ess
limit(GP, S, 0)
Ke=49.5
%Kc=K/Ke
Kc=K/Ke
```

```
%PID final
%Gc=Kp+Ki/S+Kd*S
```

```

%Gc=(Kp*S+Ki+Kd*S^2)/S
%Gc=Kd*(S^2+Kp/Kd*S+Ki/Kd)/S

%Gc=K*(S+a)*(S+b)/S
%Gc=K*(S^2+S*a+S*b+a*b)/S
%Gc=Kd*(S*S+87.32*S+4.366)/S

clear S
syms S
Gc=0.014498*(S+21.525)*(S+0.1)/S
Gc=expand(Gc)

Kd1=0.01691
Kp1=0.3657
Ki1=0.036399

%Kd*X=0.0638
%X=0.3657/0.016910
Kp=21.626
Ki=2.1525
Kd=0.014498

Gc1=Kp+'s'*Kd+Ki/'s'
%Gc2=tf([Kd Kp Ki],[1 0])

Gc1=tf([Kd Kp Ki],[1 0])
Gcl1=Gc1*Gp1/(1+Gc1*Gp1)
figure(3)
    hold on
step(Gcl1,0.1)

figure(3)
    hold off
title ("Step Temperature");
xlabel ("Time");
ylabel ("Temperature");
%text (pi, 0.7, "arbitrary text");
legend ("Signal temperature To");

%ajuste erros stado stable ess
%rampa S*GP
limit(S*GP, S, 0)
ts=4/(wn*E)
ts=1.9820E-3

```

```
figure(4)
subplot (2, 1, 1)
step(Gp1);
title ("Step Temperature");
xlabel ("Time");
ylabel ("Temperature");
%text (pi, 0.7, "arbitrary text");
legend ("Signal temperature To");
```

```
%step(feedback(Gp1))
%title ("Step Temperature");
%xlabel ("Time");
%ylabel ("Temperature");
%text (pi, 0.n7, "arbitrary text");
legend ("Signal Temperature To");
```

```
subplot (2, 1, 2)
step(Gcl1,0.1)
title ("Step Temperature");
xlabel ("Time");
ylabel ("Temperature");
%text (pi, 0.7, "arbitrary text");
legend ("Signal temperature To");
```

```
%step(feedback(Gp1))
%title ("Step Temperature");
%xlabel ("Time");
%ylabel ("Temperature");
%text (pi, 0.n7, "arbitrary text");
legend ("Signal Temperature To");
```

```
hold off
```

P.i.d Control

Engineer
Jhoberg Quevedo

$$G(s) = \frac{102339 \cdot s + 1900}{2 \cdot s \cdot (247 \cdot s^2 + 50)}$$

$$\text{overshoot} = 0,001 \text{ seg}$$

$$t_r = 0,001 \text{ tiempo levantamiento}$$

$$e_{ss} = 0,01$$

$$\textcircled{1} \quad \phi = e^{-\pi \cdot \xi / \sqrt{1-\xi^2}} \rightarrow \xi \rightarrow$$

$$\text{overshoot} = 0,01 = e^{-\pi \cdot \xi / \sqrt{1-\xi^2}} \rightarrow \xi = 0,9102$$

$$\textcircled{2} \quad \frac{\omega_n}{t_r} = 0,01 = (0,8 + 2,5 \cdot \xi) / \omega_n \rightarrow \omega_n = 3075,4$$

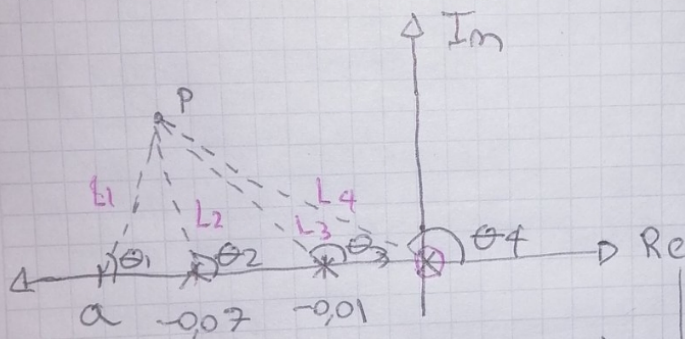
$\textcircled{3}$ root locus

roots $G(s)$

$$p : -0,066934$$

$$z : 0$$

$$z : -0,01856$$



$$\begin{aligned} s_{n1} &= -2814,3 + i1273,9 \\ s_{n1} &= -2799,2 + i1273,9 \end{aligned}$$

$$\theta_4 = 180 - \tan^{-1} \left(\frac{1273,9}{2799,2 - 0} \right) = 179,57$$

$$\theta_3 = 180 - \tan^{-1} \left(\frac{1273,9}{2799,2 - 0,01856} \right) = 179,57$$

$$\theta_2 = 180 - \tan^{-1} \left(\frac{1273,9}{2799,2 - 0,066934} \right) = 179,57$$

$$\theta_1 = \tan^{-1} \left(\frac{1273,9}{a - 2799,2} \right)$$

$$a = 21,525$$

$$\sum \theta_z - \sum \theta_p = -180$$

$$(\theta_1 + \theta_3) - (\theta_2 + \theta_4) = -180$$

$$\theta_1 - 179,57 - 179,57 - 179,57 = -180$$

$$\theta = -180 + 179,57 = -0,43 \rightarrow \frac{1273,9}{a - 2799,2} = \tan(-0,43)$$

$$a = \frac{1273,9}{\tan(-0,43)} + 2799,2 = 21,525$$

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Pi.d control

$$a = 21.525$$

(4) calculo K

$$K = \left| \frac{1}{G(s)} \cdot \frac{1}{G_c(s)} \right|, G_c(s) = K(s + 21.525)$$

$$K = \left| \frac{2.5(747.5 + 50)}{10239.5 + 1900} \right| \cdot \left| \frac{1}{(s + 21.525)} \right|_{s=|s_{n1}|}$$

$$K = 0.014498$$

$$|s_{n1}| = 3082.5$$

(5) controller P.I.D

$$G_c = \frac{0.014498 \cdot (s + 21.525) \cdot (s + b)}{s}$$

$$G_c = s \cdot 0.014498 \cdot 1 + 0.3135 + \boxed{b = 0.1}$$

$$0.031214$$

$$G_c =$$

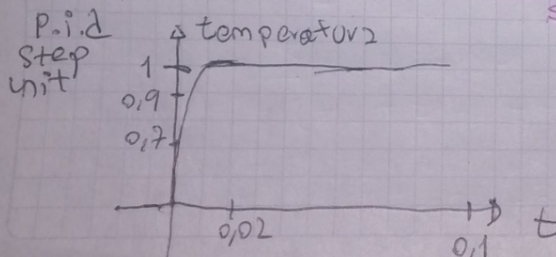
$$\rightarrow K_d \cdot X = 0.3135 \rightarrow X = 0.3135 / 0.014498$$

$$K_P = X = 21.624$$

$$K_i = 0.031214 / 0.014498 = 2.1530$$

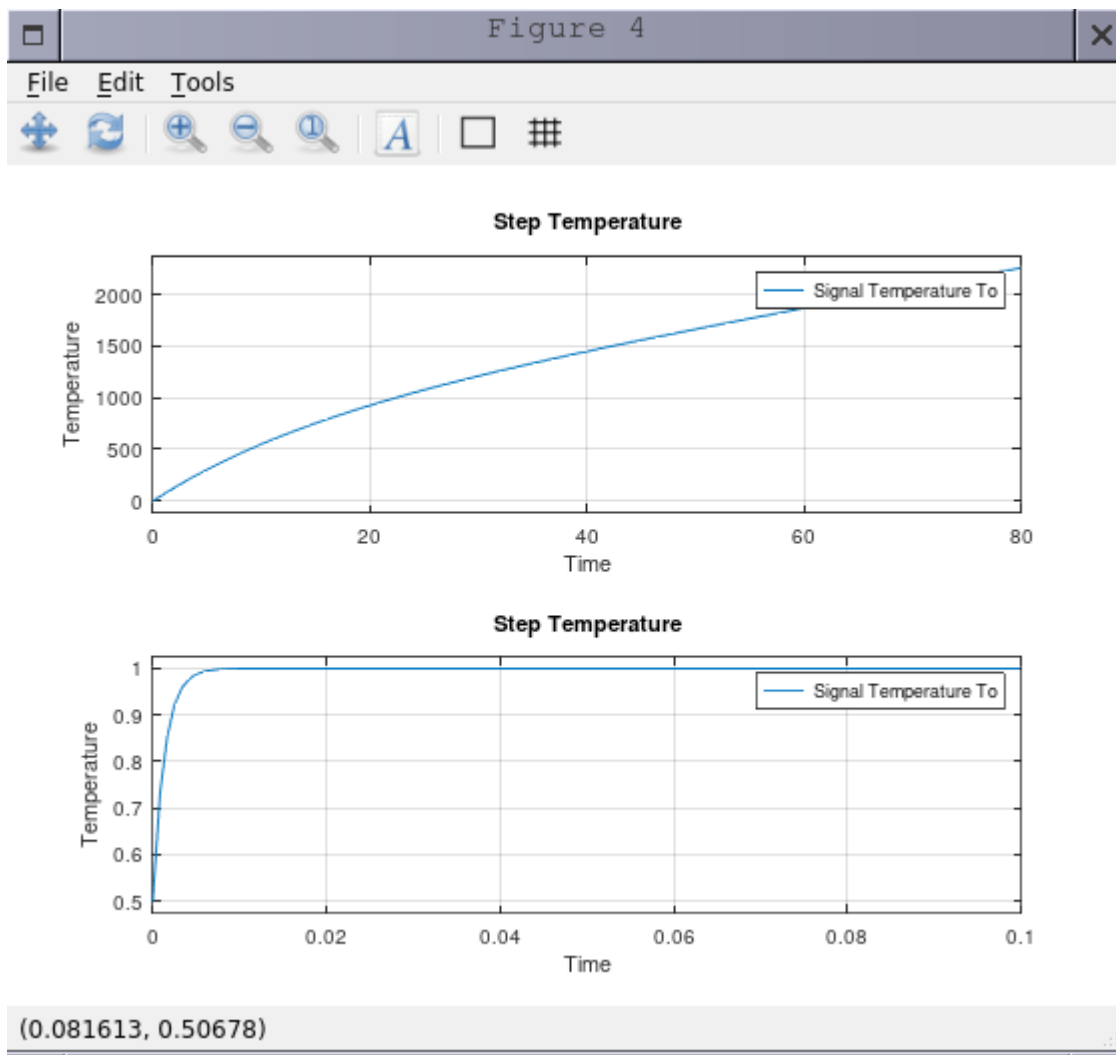
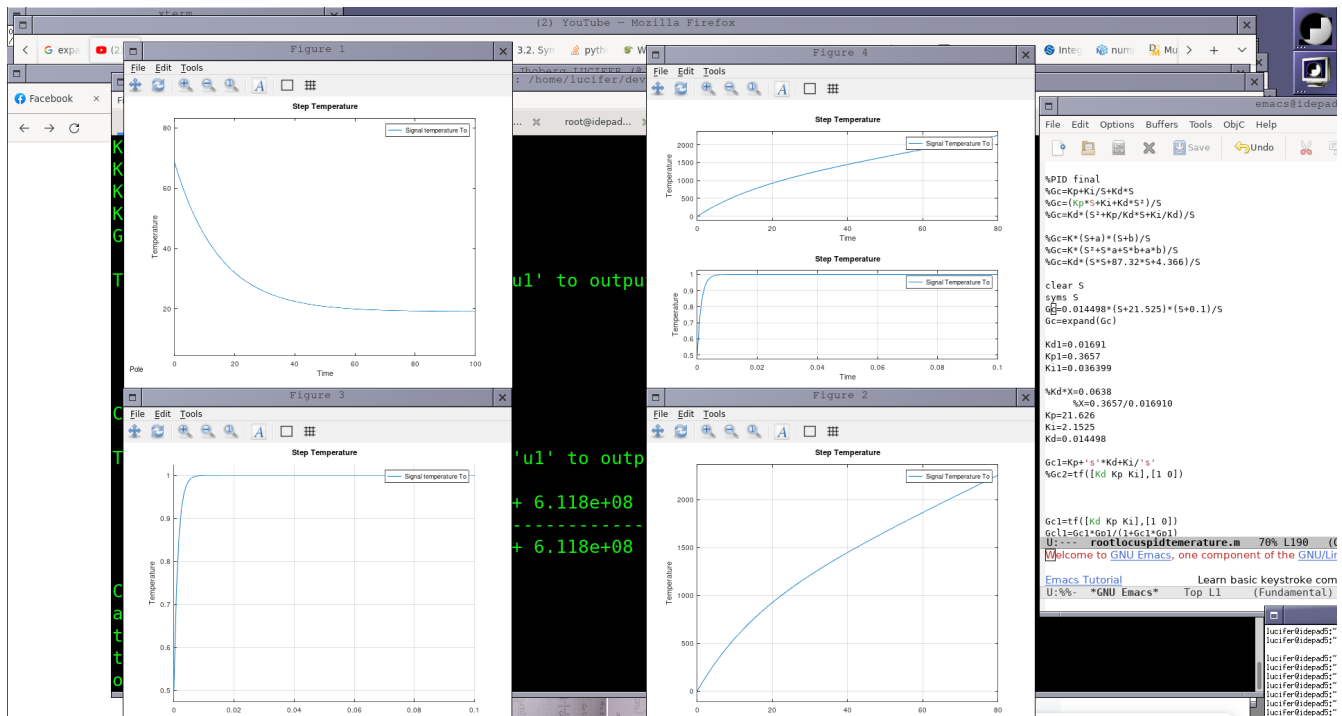
$$K_d = 0.014498$$

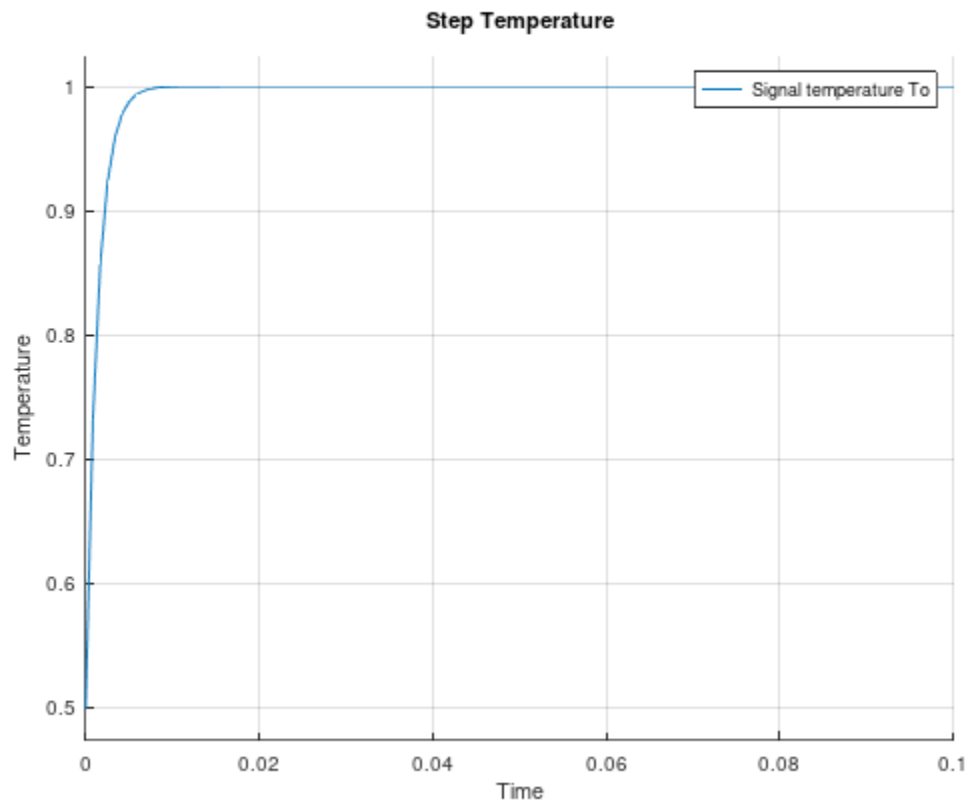
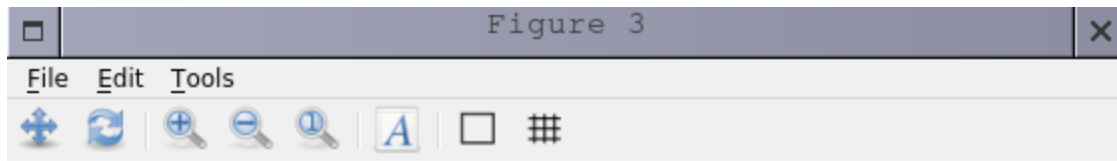
$$\underline{G_c = 0.014498 \cdot s^2 + 21.624 \cdot s + 2.1530}$$



$$G_{cl} = \frac{G(s) \cdot G_c(s)}{1 + G(s) \cdot G_c(s)}$$

step(G_{cl})





(0.08, 1.0017)



